Graphs and characteristics of quadra[redacted]nctions

1 Consider the equation $y=25-(x+2)^2$.

a Does y have a minimum or maximum value?

b State the maximum value of y.

2 The maximum value of the function $y=-7x^2-140x+m$ is $y=3$.

a Find the value of x at the point where the maximum value of the function occurs.

b Find the value of m.

3 Consider the function $f(x)=x^2-8x$.

a Does the function have any maxima or minima?

b At what value of x is $f(x)$ at its minimum?

c State the minimum value of the function.

d On which interval is the function increasing?

e On which interval is the function decreasing?

4 Consider the function $f(x)=x^2+4x+7$.

a Sketch the graph of the function.

b Does the function have any maxima or minima?

c State the minimum value of the function.

d Over which interval is the function increasing?

e Over which interval is the function decreasing?

5 State whether the following descriptions could represent the graph of $f(x)=9x^2(x+k)-x-k$, given that k is an integer.

a As $x\to-\infty$, $f(x)\to\infty$, as $x\to\infty$, $f(x)\to\infty$, and the graph has $3x$-intercepts.

b As $x\to-\infty$, $f(x)\to\infty$, as $x\to\infty$, $f(x)\to\infty$, and the graph has $4x$-intercepts.

c As $x\to-\infty$, $f(x)\to-\infty$, as $x\to\infty$, $f(x)\to\infty$, and the graph has $4x$-intercepts.

d As $x\to-\infty$, $f(x)\to-\infty$, as $x\to\infty$, $f(x)\to\infty$, and the graph has $3x$-intercepts.

6 Consider the general quadratic equation $y=ax^2+bx+c$, $a\neq0$:

a Is it correct to say that the vertex of the g[illegible] will be located at $x = \frac{b}{2a}$?

b What should the x value of the vertex actually be?

7 Consider the function $y = x^2$

a Complete the following table of values.

x	-3	-2	-1	0	1	2	3
y							

b Plot the curve.

c Are the y values ever negative?

d Write down the equation of the axis of symmetry.

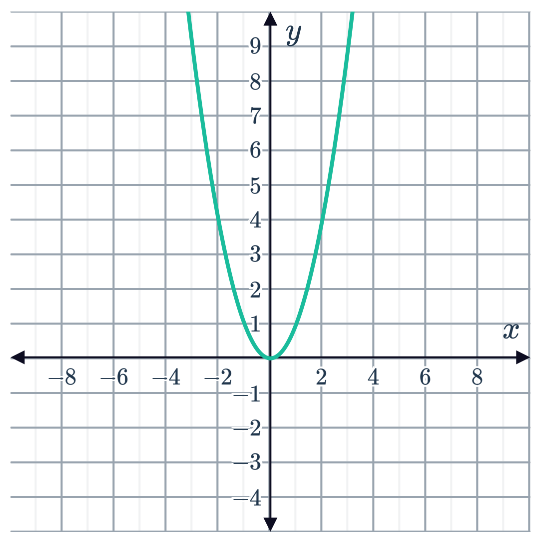
e State the minimum y value.

f For every y value greater than 0, how many corresponding x values are there?

8 Consider the graph of $y = x^2$.

a How do we shift the graph of $y = x^2$ to get the graph of $y = (x-2)^2$?

b Hence plot $y = (x-2)^2$ on the same graph as $y = x^2$.



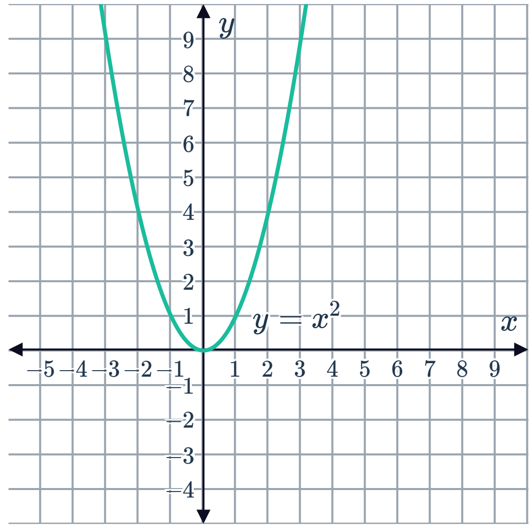
9 Consider the equation $y = x^2 + 5$.

a Complete the set of solutions for the given equation.

x	-3	-2	-1		1	2	3
y				5			

b Plot the curve that results from the entire set of solutions for the equation being

10 Consider the function $y = (x-6)^2$.



a Complete the table of values.

x	4	5	6	7	8
y					

b State the minimum value of y.

c State the x value that corresponds to the minimum y value.

d Hence state the coordinates of the vertex of the parabola.

e How many units to the right has $y = x^2$ been translated?

f Translate the parabola $y = x^2$ to sketch $y = (x-6)^2$.

11 Consider the function $y = -(x-5)^2$.

a Complete the table of values.

x	3	4	5	6	7
y					

b State the maximum value of y.

c State the x value that corresponds to the maximum y value.

d Hence state the vertex of the parabola.

e Sketch the graph of $y = -(x-5)^2$.

12 Consider the following equations:

i Find the x-value of the x-intercept.

ii Find the y-value of the y-intercept.

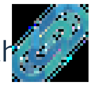
iii Determine the vertex.

iv Graph the equation.

a $y = (x-2)^2$

b $y = (x-5)^2 - 9$

13 Consider the function $y = x^2 - 6x + 5$.

a Find the x-value of the x-intercepts of thction.

b Sketch the graph of $y = x^2 - 6x + 5$.

c Plot the points corresponding to $y = 0$ on the curve.

14 Consider the curve $y = x^2 + 2x - 2$.

a State the equation of the axis of symmetry.

b Now state the minimum value of y.

c Using the minimum point on the curve, sketch the graph of the function.

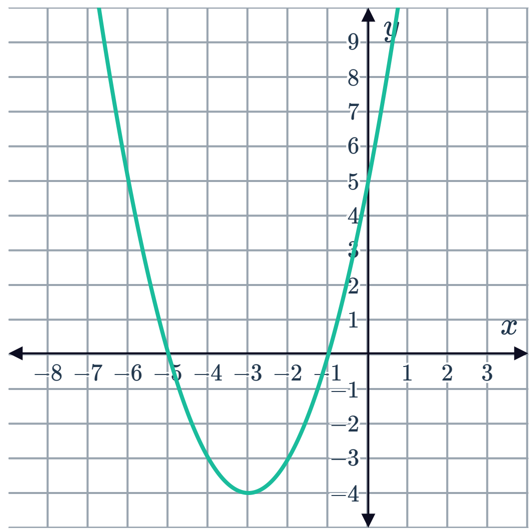
15 Consider the function $y = x^2 + 6x + 5$.

a State the equation of the axis of symmetry.

b State the minimum value of y.

c Hence state the coordinates of the vertex of the curve.

d Consider the graph of another quadratic function shown on the right. State the coordinates of its vertex.



e State whether the following statements are correct about the two graphs.

i They have the same concavity.

ii They have the same x-intercepts.

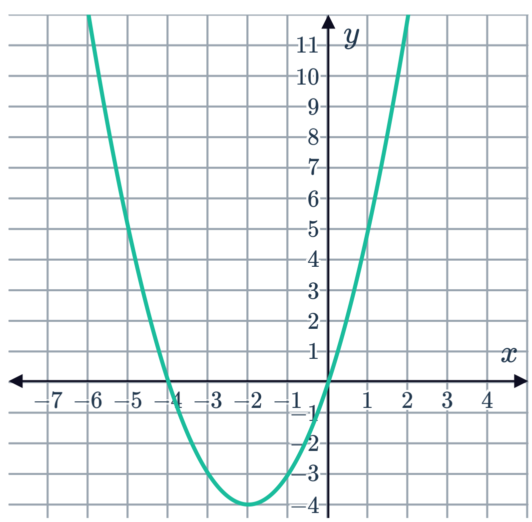
iii They share the same turning point.

16 Consider the graph of the function $y = f(x)$:

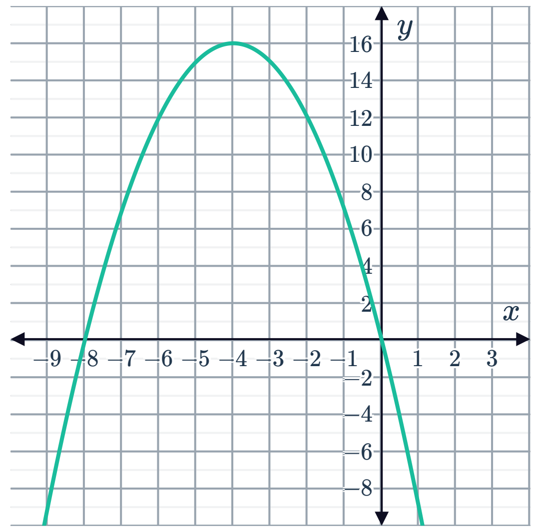
a State the y-value of the absolute minimum of the graph.

b Hence state the range of the function.

c State the domain of this function.

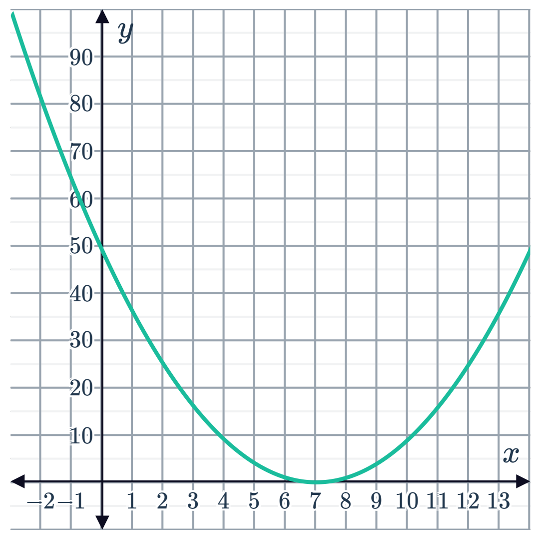


17 Consider the graph of the function $y = f(x)$:



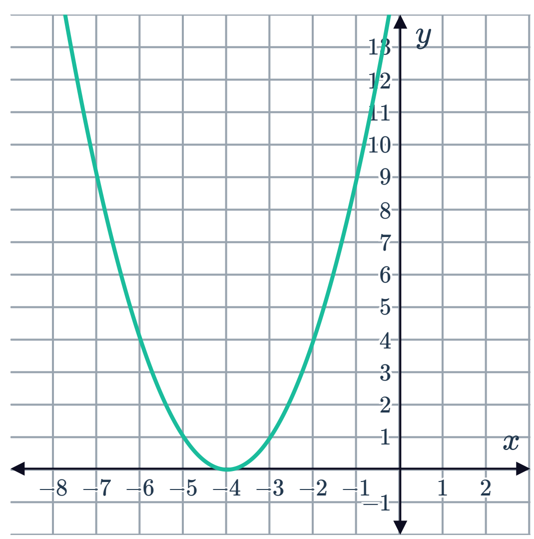
- **a** State the absolute maximum of the graph.
- **b** Hence state the range of the function.
- **c** State the domain of this function.

18 Consider the graph of the function $y = f(x)$:



- **a** State the absolute minimum of the graph.
- **b** Hence determine the range of the function.
- **c** Over what interval of the domain is the function increasing?

19 Consider the graph of the function $y = f(x)$ and answer the following questions.



- **a** State the absolute minimum of the graph.
- **b** Hence determine the range of the function.
- **c** Over what interval of the domain is the function decreasing?

20 Consider a quadratic function that passes through the points shown in the following table.

x	1	2	3	4	5	6	7
y	17	7	1	-1	1	7	17

a Does this quadratic function have a maximum or minimum point?

b State the maximum or the minimum value of the function.

c State the range of the quadratic function.

21 Consider the table of coordinates that satisfy a particular quadratic function.

x	-9	-8	-7	-6	-5	-4	-3
y	-11	-6	-3	-2	-3	-6	-11

a Does the function have a maximum or minimum point?

b State the maximum or the minimum value of the function.

c State the range of the function. Give your answer as an inequality.

22 A quadratic function includes the points shown in the table below.

x	5.5	6	6.5	7	7.5	8	8.5
y	15	5	-1	-3	-1	5	15

a Does the function have a maximum or minimum point?

b State the maximum or the minimum value of the function.

c State the range of the function.

23 Consider a quadratic function that passes through the points shown in the following table.

x	-5.5	-5	-4.5	-4	-3.5	-3	-2.5
y	-12	-7	-4	-3	-4	-7	-12

a Does the function have a maximum or minimum point?

b State the maximum or the minimum value of the function.

24 Consider the equation $y = 2x^2$.

a Is every value of y positive?

b Consider the graph of $y = 2x^2$ shown on the right. State the axis of symmetry of the parabola.

c State the minimum value of y.